

16

Extreme value theory

Syllabus objectives

- 1.4 Introduction to extreme value theory.
 - 1.4.1 Recognise extreme value distributions, suitable for modelling the distribution of severity of loss and their relationships.
 - 1.4.2 Calculate various measures of tail weight and interpret the results to compare the tail weights.

0 Introduction

In this chapter we look at how we can model extreme events. In the context of insurance, these are events that are very unlikely but can have a large financial impact. Examples include natural catastrophes such as earthquakes, man-made catastrophes such as aeroplane crashes and financial events such as stock market crashes.

Our first thought in modelling extreme events might be to fit a distribution to past data and then to use the tails of the distribution to estimate the probability of future extreme events. For example, consider a data set of past claim amounts on an insurer's motor insurance portfolio. An actuary could fit a loss distribution to these data values. However, the estimation of the parameters of the distribution would be heavily influenced by the bulk of the past claims data, which is likely to be non-extreme. Relatively little weight would be placed on the extreme data in the fitting process. Therefore, if the insurer uses the fitted distribution to estimate the probability of future extreme events, such events may be underestimated.

Better modelling of extreme events can be done by considering distributions that are fitted specifically to the tail of a dataset rather than to the entire dataset. We consider two such distributions:

1. the generalised extreme value distribution is studied in Section 2
2. the generalised Pareto distribution is studied in Section 3.

In Section 4, we look at measures of tail weight, *ie* how likely extreme values are to occur.

1 Extreme events and extreme value theory

1.1 Extreme events



Question

Define an 'extreme' event in terms of its frequency and severity.

Solution

An extreme event is one that occurs with very low frequency and very high severity.

1.2 Difficulties modelling extreme events

Low frequency events involving large losses can have a devastating impact on companies and investment funds. The 'credit crunch' that started in 2007 was an example of this. It generated more extreme movements in share prices than had been seen for over 20 years previously.

The credit crunch highlighted many deficiencies in the modelling of extreme events by financial providers. For example:

- Financial risk events such as asset price movements were often modelled using normal distributions. However, empirical evidence suggests that this is not the case in practice. We discuss this further below.
- Distributions were typically fitted to whole (rather than extreme) datasets of asset price movements. This resulted in fitted distributions that understated the probability of extreme events.
- Many financial providers focused on the losses predicted to occur at the 99th percentile or 99.5th percentile point of a distribution. Providers held capital to meet the losses at these points. Little attention was paid to the expected loss (or range of losses) beyond these points.
- The models failed to recognise that, in times of financial crisis, correlations between risk events increase. Hence the models underestimated the *joint* probability of multiple extreme risk events happening at once.

Since the credit crunch, there has been increased focus on the techniques used to model extreme events.

So it is important to ensure that we model the form of the distribution in the tails correctly. However, the low frequency of these events also means that there is relatively little data to model their effects accurately.

Hence the irony of the situation – the tail data with which to fit the distribution is sparse, yet the distribution is being used to model tail risk events.

Many types of financial data tend to be much more narrowly peaked in the centre of the distribution and to have fatter tails than the normal distribution. This shape of distribution is known as leptokurtic. For example, when share prices are modelled, large price movements occur more frequently than predicted by the normal distribution. So the normal distribution may be unsuitable for modelling the large movements in the tails.

The word 'leptokurtic' is a measure of the kurtosis of a distribution, which is the fourth *standardised* central moment of a distribution:

$$\kappa = E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right]$$

All (univariate) normal distributions have kurtosis equal to three and are described as mesokurtic. A distribution with kurtosis greater than three is leptokurtic (more peaked with fatter tails). A distribution with kurtosis less than three is platykurtic (a broader peak with more slender tails).

One reason for these fat tails is that the volatility of financial variables does not remain constant, but varies stochastically over time. This property is known as *heteroscedasticity*.

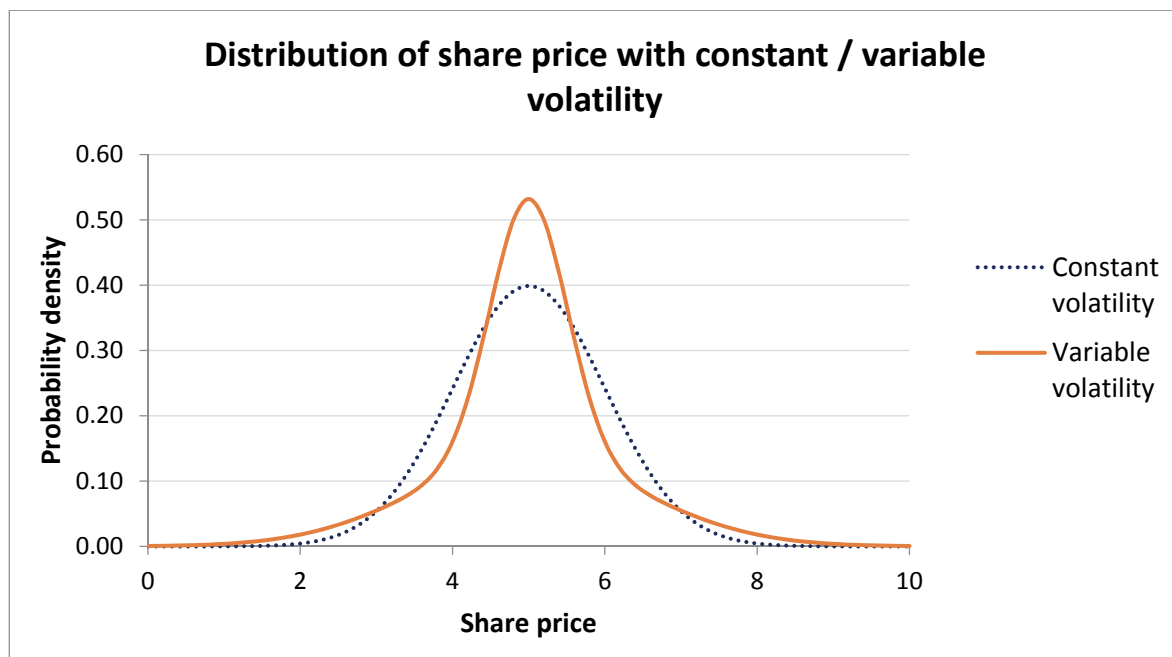
The volatility of a random variable is equivalent to its standard deviation.



Question

The graph below compares two distributions for the price of a share in one year's time:

- a $N(5, \sigma^2)$ distribution with constant volatility, $\sigma = 1$
- a $N(5, \sigma^2)$ distribution where the volatility is heteroscedastic, ie $\sigma = 0.5$ and $\sigma = 1.5$ with equal probability.



Comment on the relative shape of the graphs.

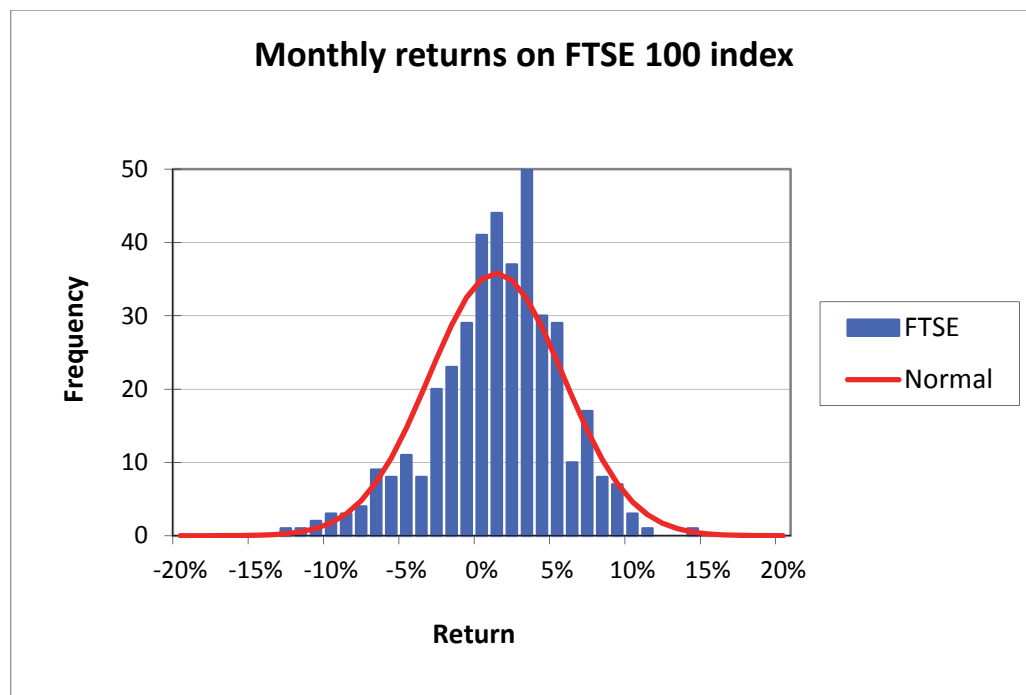
Solution

In the case where the volatility is variable, the resultant probability density function is more peaked with fatter tails, *ie* it is leptokurtic.

When empirical asset return data values are analysed, they exhibit volatility clustering, *ie* periods of sustained high volatility and periods of sustained low volatility. This suggests that volatility of asset return data is not constant but heteroscedastic.

Even if we select an appropriate form of fat-tailed distribution, if we attempt to fit the distribution using the whole of our dataset, this is unlikely to result in a good model for the tails, since the parameter estimates will be heavily influenced by the main bulk of the data in the central part of the distribution.

This is illustrated by the graph below, which shows the frequency of (log) returns on the FTSE-100 between April 1984 and September 2017. A normal distribution has been fitted to the whole dataset. It can be seen, from the graph, that the normal distribution underestimates the probability of the extreme events in the lower tail and overestimates extreme events in the upper tail.



1.4 Extreme value theory

Fortunately, better modelling of the tails of the data can be done through the application of extreme value theory. The key idea of extreme value theory is that the asymptotic behaviour of the tails of most distributions can be accurately described by certain families of distributions.



Question

Explain what is meant by the phrase ‘asymptotic behaviour of the tails of a distribution’ in the paragraph above.

Solution

The phrase is referring to how the distribution behaves in the limit, as a certain parameter (such as the number of observations in a sample) tends to infinity.

More specifically, the maximum values of a distribution (when appropriately standardised) and the values exceeding a specified threshold (called threshold exceedances) converge to two particular families of distributions as the sample size increases.

These two families of distributions are:

- generalised extreme value distributions, and
- generalised Pareto distributions.

We look at these in Sections 2 and 3 respectively.

2 Generalised extreme value (GEV) distribution

If we are dealing with losses that have typical sizes, ie ones whose values come from the central part of the distribution, we can make use of the Central Limit Theorem.

This tells us that, if we calculate the mean, $\bar{X}$, of a set of n values taken from a loss distribution that has mean μ and variance σ^2 , the standardised value, $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$, can be approximated using the standard normal distribution.

However, the most financially significant part of a loss distribution is usually the right-hand tail where the large losses occur. These are the *extreme values* of the distribution. So, is there a similar way to approximate the behaviour of the extreme values in the tail of the distribution?

The Core Reading here is referring to a distribution of losses (rather than returns or profits) and so we are concerned with the extreme right-hand tail of the loss distribution.

2.1 Maximum values

One approach is to look at $X_M = \max\{X_1, X_2, \dots, X_n\}$, the maximum value in a set of n values. This is referred to as a *block maximum*.



Question

The dataset below shows the claim amounts in £000s in respect of a commercial property portfolio over a period of a year.

Claim number	Claim amount	Claim number	Claim amount	Claim number	Claim amount	Claim number	Claim amount
1	9	17	12	33	19	49	118
2	28	18	35	34	17	50	55
3	20	19	12	35	66	51	14
4	8	20	75	36	55	52	94
5	102	21	80	37	81	53	54
6	152	22	42	38	140	54	81
7	23	23	9	39	64	55	62
8	108	24	122	40	9	56	83
9	42	25	145	41	9	57	23
10	12	26	13	42	36	58	19
11	110	27	16	43	185	59	55
12	9	28	113	44	135	60	104
13	22	29	9	45	25		
14	37	30	8	46	16		
15	147	31	12	47	55		
16	128	32	84	48	31		

- (i) Determine the values of X_M where the block size is:
- (a) $n = 5$
- (b) $n = 10$
- (ii) Comment on the trade-off between the block size and the values of X_M that will be used to fit the extreme value distribution.

Solution

- (i)(a) The values of X_M are $\{102, 152, 147, 128, 145, 113, 84, 140, 185, 118, 94, 104\}$.
- (i)(b) The values of X_M are $\{152, 147, 145, 140, 185, 104\}$.
- (ii) The larger the block size, the fewer the number of blocks (eg when $n = 10$ there are six blocks whereas when $n = 5$ there are twelve blocks). The fewer the number of blocks, the fewer (but more 'extreme') the values of X_M that will be used to fit the extreme value distribution.
-

If we look at a number of such blocks, we find that these maximum values can be standardised in a similar way, ie we can calculate expressions of the form $\frac{X_M - \alpha_n}{\beta_n}$ that can be approximated by a particular type of distribution – called an **extreme value distribution**.

As with the Central Limit Theorem, we are interested in determining the distribution of the standardised quantity:

$$\frac{X_M - \alpha_n}{\beta_n}$$

where $X_M = \max\{X_1, X_2, \dots, X_n\}$ with each X_i representing an observed loss. The α_n and β_n are appropriately chosen constants. We will give an example of these in the next section.

2.2 Distribution of the (standardised) maximum values

In order to determine the distribution of this standardised quantity, we consider its CDF.

If the values are independent and identically distributed (IID), each with cumulative distribution function, $F(x)$, the cumulative distribution function of the block maximum is:

$$\begin{aligned} P(X_M \leq x) &= P(X_1 \leq x, X_2 \leq x, \dots, X_n \leq x) \\ &= P(X_1 \leq x)P(X_2 \leq x) \dots P(X_n \leq x) \\ &= [P(X \leq x)]^n \\ &= [F(x)]^n \end{aligned}$$

We can attempt to standardise the values of X_M by finding a sequence of constants $\alpha_1, \alpha_2, \dots$ and $\beta_1, \beta_2, \dots > 0$ so that the limiting distribution:

$$\lim_{n \rightarrow \infty} P\left(\frac{X_M - \alpha_n}{\beta_n} \leq x\right) = \lim_{n \rightarrow \infty} [F(\beta_n x + \alpha_n)]^n$$

depends only on x .

The Extreme Value Theorem tells us that it is possible to find such values of α_n and β_n for most common distributions.

For example, if the individual losses are distributed exponentially with $F(x) = 1 - e^{-\lambda x}$, we can set $\alpha_n = \frac{1}{\lambda} \ln n$ and $\beta_n = \frac{1}{\lambda}$.

We don't need to concern ourselves with how these values for α_n and β_n are chosen for the exponential distribution, only that they can be chosen.



Question

Let $X \sim \text{Exp}(\lambda)$, $\alpha_n = \frac{1}{\lambda} \ln n$ and $\beta_n = \frac{1}{\lambda}$ for all n .

By substituting in for α_n and β_n and by using the CDF of the exponential distribution, determine the CDF of the limiting distribution of the *standardised* values of X_M :

$$\lim_{n \rightarrow \infty} P\left(\frac{X_M - \alpha_n}{\beta_n} \leq x\right) = \lim_{n \rightarrow \infty} [F(\beta_n x + \alpha_n)]^n$$

(Hint: $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$.)

Solution

(i) The solution below forms part of the Core Reading.

$$\begin{aligned} \lim_{n \rightarrow \infty} [F(\beta_n x + \alpha_n)]^n &= \lim_{n \rightarrow \infty} \left[F\left(\frac{1}{\lambda} x + \frac{1}{\lambda} \ln n\right) \right]^n \\ &= \lim_{n \rightarrow \infty} \left\{ 1 - \exp\left[-\lambda\left(\frac{1}{\lambda} x + \frac{1}{\lambda} \ln n\right)\right] \right\}^n \\ &= \lim_{n \rightarrow \infty} \left\{ 1 - \exp(-x - \ln n) \right\}^n \\ &= \lim_{n \rightarrow \infty} \left\{ 1 - \frac{e^{-x}}{n} \right\}^n = e^{-e^{-x}} \end{aligned}$$

The last line in the Core Reading above follows from the hint:

$$\lim_{n \rightarrow \infty} \left\{ 1 - \frac{e^{-x}}{n} \right\}^n = \lim_{n \rightarrow \infty} \left\{ 1 + \left(\frac{-e^{-x}}{n} \right) \right\}^n = e^{-e^{-x}}$$

This distribution is known as the standard Gumbel distribution.

The standard Gumbel distribution is a particular type of extreme value distribution, which we consider in more detail later on.

2.3 GEV distribution

More generally, whatever the underlying distribution of the data, the distribution of the standardised maximum values will converge to a distribution called the *generalised extreme value* (GEV) distribution as n increases, ie $\lim_{n \rightarrow \infty} [F(\beta_n x + \alpha_n)]^n = H(x)$.

In Section 2.2, we considered the specific case where the underlying distribution is $X \sim \text{Exp}(\lambda)$. However, the generalised extreme value distribution is more generic than this and caters for a number of different underlying distributions for X .

The cumulative distribution function of the GEV distribution is:

$$H(x) = \begin{cases} \exp \left(- \left(1 + \frac{\gamma(x - \alpha)}{\beta} \right)^{-\frac{1}{\gamma}} \right) & \gamma \neq 0 \\ \exp \left(- \exp \left(- \frac{(x - \alpha)}{\beta} \right) \right) & \gamma = 0 \end{cases}$$

This distribution has three parameters:

- a location parameter α
- a scale parameter $\beta > 0$
- a shape parameter γ .

The parameters α and β just rescale (shift and stretch) the distribution. They are analogous to (but do not usually correspond to) the mean and standard deviation.

The parameter γ determines the overall shape of the distribution (analogous to the skewness) and its sign (positive, negative or zero) results in three different shaped distributions.



Question

Derive the PDF for the GEV distribution.

Solution

In the case where $\gamma \neq 0$:

$$H(x) = \exp \left(- \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\frac{1}{\gamma}} \right)$$

We must differentiate this expression to obtain the PDF. Let $u = \frac{\gamma(x-\alpha)}{\beta}$ and $v = (1+u)^{-\frac{1}{\gamma}}$ so that $H(x) = \exp(-v)$. Then:

$$\frac{du}{dx} = \frac{\gamma}{\beta}$$

$$\frac{dv}{du} = \left(-\frac{1}{\gamma} \right) (1+u)^{-\frac{1}{\gamma}-1} = -\frac{1}{\gamma} \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\left(1+\frac{1}{\gamma}\right)}$$

$$\text{and } \frac{dH(x)}{dv} = -\exp(-v) = -\exp \left(- (1+u)^{-\frac{1}{\gamma}} \right) = -\exp \left(- \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\frac{1}{\gamma}} \right).$$

So the PDF is:

$$\begin{aligned} h(x) &= \frac{dH(x)}{dx} \\ &= \frac{dH(x)}{dv} \times \frac{dv}{du} \times \frac{du}{dx} \\ &= -\exp \left(- \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\frac{1}{\gamma}} \right) \times -\frac{1}{\gamma} \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\left(1+\frac{1}{\gamma}\right)} \times \frac{\gamma}{\beta} \\ &= \frac{1}{\beta} \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\left(1+\frac{1}{\gamma}\right)} \exp \left(- \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\frac{1}{\gamma}} \right) \end{aligned}$$

In the case where $\gamma=0$:

$$H(x) = \exp \left(-\exp \left(-\frac{(x-\alpha)}{\beta} \right) \right)$$

Again, we must differentiate the above expression to obtain the PDF. Let $u = \frac{x-\alpha}{\beta}$ and $v = \exp(-u)$ so that $H(x) = \exp(-v)$. Then:

$$\frac{du}{dx} = \frac{1}{\beta}$$

$$\frac{dv}{du} = -\exp(-u) = -\exp\left(-\frac{(x-\alpha)}{\beta}\right)$$

and $\frac{dH(x)}{dv} = -\exp(-v) = -\exp(-\exp(-u)) = -\exp\left(-\exp\left(-\frac{(x-\alpha)}{\beta}\right)\right).$

So the PDF is:

$$\begin{aligned} h(x) &= \frac{dH(x)}{dx} \\ &= \frac{dH(x)}{dv} \times \frac{dv}{du} \times \frac{du}{dx} \\ &= -\exp\left(-\exp\left(-\frac{(x-\alpha)}{\beta}\right)\right) \times -\exp\left(-\frac{(x-\alpha)}{\beta}\right) \times \frac{1}{\beta} \\ &= \frac{1}{\beta} \exp\left(-\left[\frac{(x-\alpha)}{\beta} + \exp\left(-\frac{(x-\alpha)}{\beta}\right)\right]\right) \end{aligned}$$

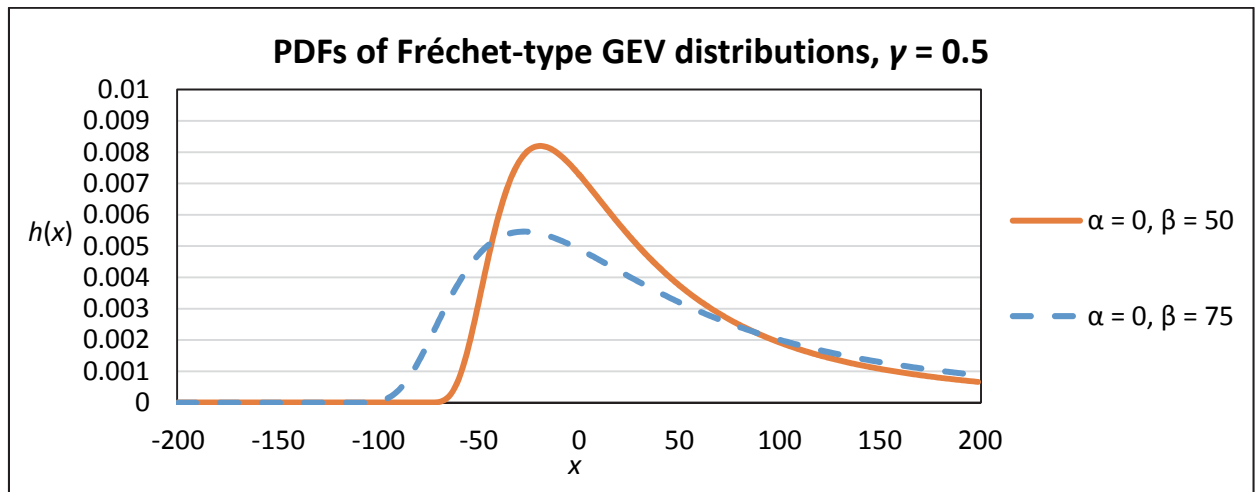
2.4 Fréchet, Weibull and Gumbel GEV distributions

The GEV family of distributions subdivides into three distinct classes depending on the value of the shape parameter, γ . These are called:

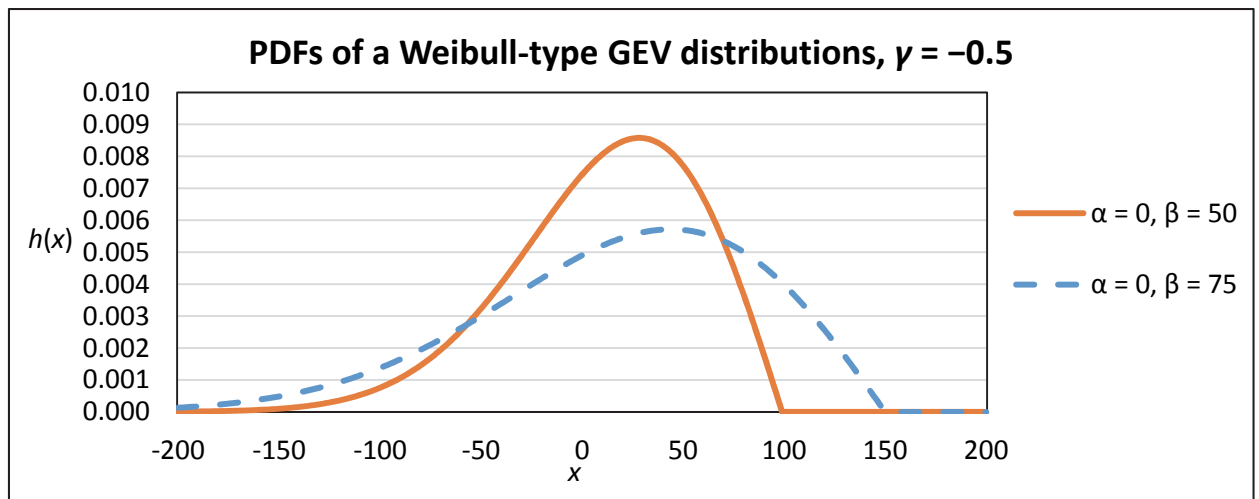
- Fréchet-type GEV distributions – when $\gamma > 0$
- Weibull-type GEV distributions – when $\gamma < 0$
- Gumbel-type GEV distributions – when $\gamma = 0$.

Each type gives rise to a different shape of PDF as can be seen in the three graphs below.

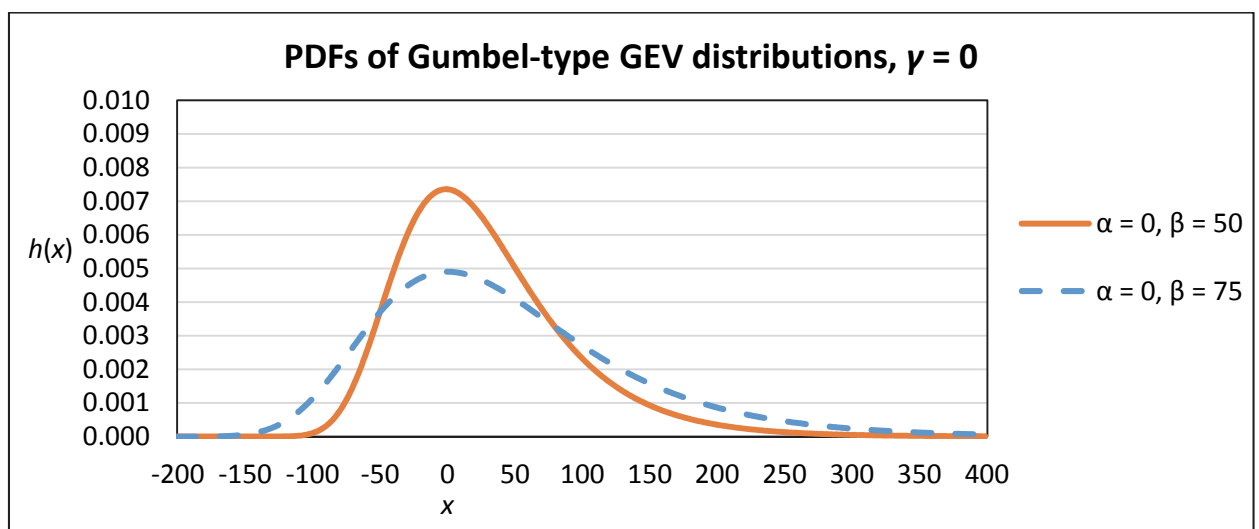
Fréchet-type GEV distributions



Weibull-type GEV distribution



Gumbel-type GEV distributions





Question

Comment briefly on the range of values that x takes for each of the Fréchet-type, Weibull-type and Gumbel-type GEV distributions, with reference to the three graphs above.

Solution

- When $\gamma > 0$, x is bounded below and when $\gamma < 0$, x is bounded above.
- When $\gamma = 0$, x is unbounded.

We will now explain algebraically how these bounds arise.

Fréchet-type GEV distribution

For $\gamma > 0$, the distribution is a Fréchet-type GEV distribution. Earlier, we derived the PDF as:

$$h(x) = \frac{1}{\beta} \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\left(1+\frac{1}{\gamma}\right)} \exp \left(- \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\frac{1}{\gamma}} \right)$$

In order for the PDF to be non-negative, we require the expression $\left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)$ to be positive.

(The other factors in the PDF are positive since $\beta > 0$ and the exponential function takes positive values only.)

Since $\gamma > 0$, this results in a lower bound on the values that x can take:

$$1 + \frac{\gamma(x-\alpha)}{\beta} > 0 \Rightarrow x - \alpha > \frac{-\beta}{\gamma} \Rightarrow x > \alpha - \frac{\beta}{\gamma}$$

The Fréchet-type GEV distributions tend to be those most suitable for modelling extreme financial (loss) events. This is because there is no upper bound to the loss events but also because of the tail of the distribution. Fréchet-type GEV distributions have a heavier tail (*ie* a tail that decays more slowly to 0) than other types of GEV distribution. This is due to the behaviour of the PDF:

- The factor $\exp \left(- \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\frac{1}{\gamma}} \right)$ tends to 1 as $x \rightarrow \infty$.
- The factor $\frac{1}{\beta} \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\left(1+\frac{1}{\gamma}\right)}$ tends to 0 as $x \rightarrow \infty$. It decays in accordance with what is known as the negative power law, *ie* in proportion to x^{-k} where $k > 0$.

Weibull-type GEV distribution

For $\gamma < 0$, the distribution is a Weibull-type GEV distribution. The PDF is of the same form as the Fréchet-type GEV distribution, *ie* it is given by:

$$h(x) = \frac{1}{\beta} \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\left(1+\frac{1}{\gamma}\right)} \exp \left(- \left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)^{-\frac{1}{\gamma}} \right)$$



Question

Determine a bound for values of x for the Weibull-type GEV distribution.

Solution

In order for PDF to be positive, we require the factor $\left(1 + \frac{\gamma(x-\alpha)}{\beta} \right)$ to be positive. Since $\gamma < 0$, this results in an upper bound on the values that x can take:

$$1 + \frac{\gamma(x-\alpha)}{\beta} > 0 \Rightarrow \frac{\gamma(x-\alpha)}{\beta} > -1 \Rightarrow x - \alpha < \frac{-\beta}{\gamma} \Rightarrow x < \alpha - \frac{\beta}{\gamma}$$

We might expect to fit such a distribution to, for example, the ages of a human population (indicating an upper bound to possible age) or where a loss is certain not to exceed a certain value (for example, if such losses are reinsured).

Gumbel-type GEV distribution

When $\gamma = 0$, the GEV distribution reduces to the Gumbel distribution.

In this case, the PDF is given by:

$$h(x) = \frac{1}{\beta} \exp \left(- \left[\frac{(x-\alpha)}{\beta} + \exp \left(- \frac{(x-\alpha)}{\beta} \right) \right] \right)$$

This has a tail that decays exponentially. The decay is more rapid (*ie* the tail is lighter) than that for the Fréchet-type GEV distribution given the same values of α and β .

If $\alpha = 0$ and $\beta = 1$ this becomes the PDF of the standard Gumbel distribution (whose CDF we saw in the example in Section 2.2).

The standard Gumbel distribution is the extreme value distribution arising from an exponential distribution.

2.5 Choosing the form of the GEV distribution

If we know the form of the underlying distribution, it is possible to work out the limiting distribution of the maximum value. We can then use the appropriate member of the GEV family to model the tail of the distribution.

Note the distinction between the distribution that applies to the full dataset (the underlying distribution) and the distribution that we are using to model the extreme values (the GEV distribution). Different underlying distributions are closely related to different GEV distributions. The relationship depends on factors such as whether the underlying distribution has a finite end point (*ie* whether there is a lower or upper bound to the values that can be taken by the underlying random variable), and whether the tail of the PDF of the underlying distribution exhibits exponential or power decay.

The underlying distribution will determine which of the three different types of GEV distribution will arise, as indicated in the table below. The three types are distinguished by the sign of the shape parameter γ and are named after their original discoverers.

	GEV distributions (for the maximum value) corresponding to common loss distributions		
Type	WEIBULL	GUMBEL	FRÉCHET
Shape parameter	$\gamma < 0$	$\gamma = 0$	$\gamma > 0$
Underlying distribution	Beta Uniform Triangular	Chi-square Exponential Gamma Lognormal Normal Weibull	Burr F Log-gamma* Pareto t
Range of values permitted	$x < \alpha - \frac{\beta}{\gamma}$	$-\infty < x < \infty$	$x > \alpha - \frac{\beta}{\gamma}$

* Note that $X \sim \text{loggamma}$ if $\ln X \sim \text{Gamma}$.

Unhelpfully, the extreme value distribution corresponding to the Weibull distribution from the *Tables* is actually of the Gumbel type (rather than the Weibull type).

Mathematicians have determined criteria that can be used to predict which family a particular distribution belongs to. As a rough guide:

- underlying distributions that have finite upper limits (eg the uniform distribution) are of the Weibull type (which also has a finite upper limit).
- 'light tail' distributions that have finite moments of all orders (eg exponential, normal, lognormal) are typically of the Gumbel type
- 'heavy tail' distributions whose higher moments can be infinite are of the Fréchet type.



For example, suppose we have monthly claim data stored in a matrix `data` with the first column `month` and the second column `claim`.

To calculate the block maxima for these claims using block sizes of 12 months, we would use the following R code:

```
block<-(data-1)%/%12+1
blockmax<-aggregate(claim~block,data,max)
```

We can plot a histogram of the block maxima using the `hist` function and an empirical density function using `density` in the `plot` function (if there is enough data). We can then superimpose a GEV distribution to see if it is a good approximation.

```
GEV <- function(z) {1/beta*(1+gamma*(z-alpha)/beta)^-
  (1+1/gamma)*exp(-(1+gamma*(z-alpha)/beta)^(-1/gamma)) }
lines(<sequence of x values>,GEV(<sequence of x values>))
```

The `qqplot` function is used to compare the sample data to simulated values from a fitted GEV model.

We can estimate the maximum likelihood values as we did in [Chapter 15](#) by defining the log-likelihood function and using the function `nlm` on the negative value of the log-likelihood function as before.



Question

In the question in Section 2.1, the block maximum, X_M took the values:

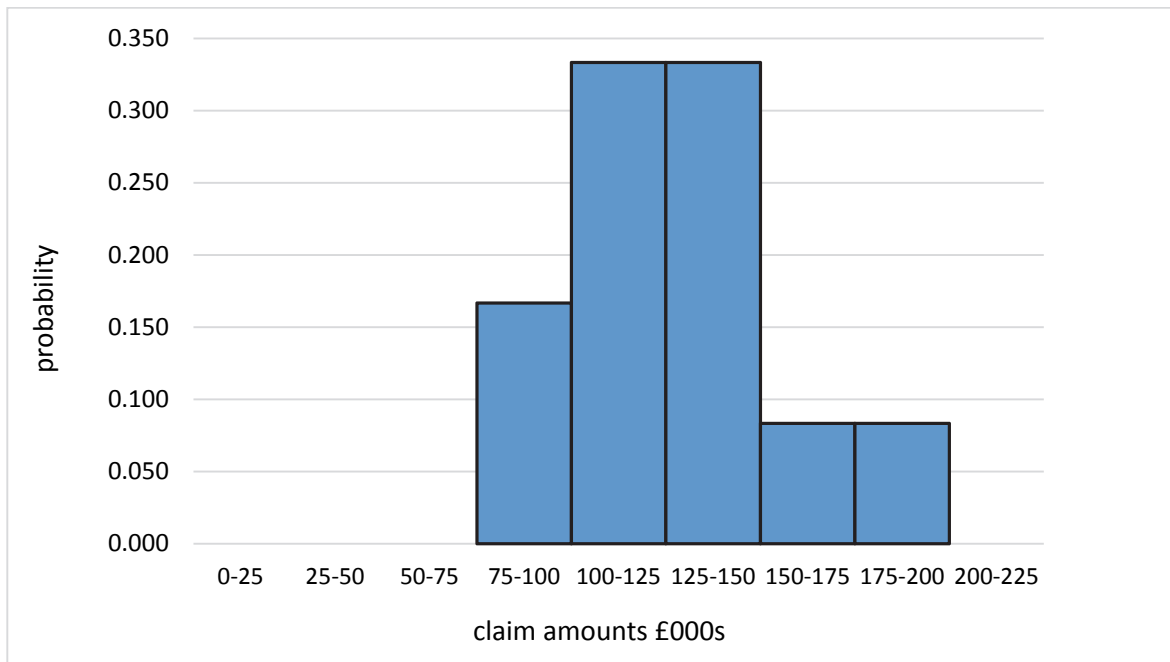
$$\{102, 152, 147, 128, 145, 113, 84, 140, 185, 118, 94, 104\}$$

when the block size was 5.

Plot these points on a frequency diagram and suggest a type of GEV distribution that might be appropriate.

Solution

A frequency diagram representing the above block maximum data is given below:



The data set is too small to be able to suggest a type of GEV distribution with any confidence. However, the upper tail decay appears to be rapid, which might lead us to consider a Gumbel-type GEV distribution.

3 Generalised Pareto distribution (GPD)

3.1 Threshold exceedances

As an alternative to focusing on the maximum value, we can consider the distribution of all the values of the variable that exceed some (large) specified threshold, eg all claims exceeding £1 million. For large samples, the distribution of these extreme values converges to the *generalised Pareto distribution*. This enables us to model the tail of a distribution by selecting a suitably high threshold and then fitting a generalised Pareto distribution to the observed values in excess of that threshold.

For example, under excess of loss reinsurance, this distribution could be used to model the claim amounts above a retention limit, ie the amounts that will pass to the reinsurer. Excess of loss reinsurance is studied in detail in [Chapter 18](#).

If we let X be a random variable with cumulative distribution function, F , then the excess over the threshold, u , is $X - u \mid X > u$.



Question

The dataset below shows the claim amounts in £000s in respect of a commercial property portfolio over a period of a year. (This is the dataset from the question in Section 2.1.)

Claim number	Claim amount	Claim number	Claim amount	Claim number	Claim amount	Claim number	Claim amount
1	9	17	12	33	19	49	118
2	28	18	35	34	17	50	55
3	20	19	12	35	66	51	14
4	8	20	75	36	55	52	94
5	102	21	80	37	81	53	54
6	152	22	42	38	140	54	81
7	23	23	9	39	64	55	62
8	108	24	122	40	9	56	83
9	42	25	145	41	9	57	23
10	12	26	13	42	36	58	19
11	110	27	16	43	185	59	55
12	9	28	113	44	135	60	104
13	22	29	9	45	25		
14	37	30	8	46	16		
15	147	31	12	47	55		
16	128	32	84	48	31		

(i) Calculate the values of $X - u \mid X > u$ when:

(a) $u = 100$

(b) $u = 125$

- (ii) Comment on the trade-off in the choice of the threshold, u .

Solution

- (i)(a) The values of $X - 100 | X > 100$ are $\{2, 52, 8, 10, 47, 28, 22, 45, 13, 40, 85, 35, 18, 4\}$.
- (i)(b) The values of $X - 125 | X > 125$ are $\{27, 22, 3, 20, 15, 60, 10\}$.
- (ii) The higher the value of the threshold, the more extreme the values of X . However, using a higher threshold means that we have fewer values with which to fit the extreme value distribution.
-



For example, suppose we have monthly claim data stored in a matrix `data` with the first column `month` and the second column `claim`.

To calculate the threshold exceedances, x_e , for these claims, at the threshold u we would use the following R code:

```
x<-data[, -1]
xe<-x[x>u]-u
```

Sometimes, the value of the threshold, u , may be specified, *eg* if it is a reinsurance retention limit. Other times, we may need to make a judgement as to where the threshold should be. Typically we may choose the threshold to be say the 90th or 95th percentile of the underlying distribution. The choice of u also depends on there being a sufficient volume of data available above the selected threshold.

If the maximum possible value of X is $x_F \leq \infty$, the cumulative distribution function of the excess is (for $0 \leq x < x_F - u$):

$$\begin{aligned}
 F_u(x) &= P(X - u \leq x | X > u) = \frac{P(X - u \leq x, X > u)}{P(X > u)} \\
 &= \frac{P(X \leq x + u, X > u)}{P(X > u)} \\
 &= \frac{P(X \leq x + u) - P(X \leq u)}{P(X > u)} \\
 &= \frac{F(x + u) - F(u)}{1 - F(u)}
 \end{aligned}$$

For example, if the individual losses are distributed exponentially with $F(x) = 1 - e^{-\lambda x}$, we have:

$$F_u(x) = \frac{\{1 - e^{-\lambda(x+u)}\} - \{1 - e^{-\lambda u}\}}{1 - \{1 - e^{-\lambda u}\}} = \frac{e^{-\lambda u} - e^{-\lambda(x+u)}}{e^{-\lambda u}} = 1 - e^{-\lambda x}$$

So, in this case, the threshold exceedances follow the same exponential distribution as X , irrespective of the threshold applied.

This is the memoryless property of the exponential distribution.



Question

With reference to the question above, use the method of maximum likelihood to fit a distribution to the threshold exceedances when the underlying claims distribution is exponential and the threshold is chosen to be 100.

Solution

The values of $X - 100 | X > 100$ are $\{2, 52, 8, 10, 47, 28, 22, 45, 13, 40, 85, 35, 18, 4\}$.

The result in the Core Reading above tells us that, if $X \sim \text{Exp}(\lambda)$ then $W = X - u | X > u \sim \text{Exp}(\lambda)$.

The likelihood function is given by:

$$L(\lambda) = \prod_{i=1}^n f_W(w_i) = \prod_{i=1}^n \lambda \exp(-\lambda w_i) = \lambda^n \exp\left(-\lambda \sum_{i=1}^n w_i\right)$$

Using the data values above, we have $n = 14$ and $\sum_{i=1}^{14} w_i = 409$, so that:

$$L(\lambda) = \lambda^{14} \exp(-409\lambda)$$

Taking natural logs:

$$\ln L(\lambda) = 14 \ln \lambda - 409\lambda$$

Differentiating with respect to λ :

$$\frac{d \ln L(\lambda)}{d\lambda} = \frac{14}{\lambda} - 409$$

This is equal to 0 when:

$$\lambda = \frac{14}{409}$$

Differentiating a second time

$$\frac{d^2 \ln L(\lambda)}{d\lambda^2} = -\frac{14}{\lambda^2} < 0 \Rightarrow \max$$

So the maximum likelihood estimate of λ is $\frac{14}{409}$ or 0.0342, and hence the fitted distribution for the threshold exceedances is $\text{Exp}(0.0342)$.

3.2 Generalised Pareto distribution

More generally we find that, whatever the underlying distribution of the data, the distribution of the threshold exceedances will converge to a *generalised Pareto distribution* as the threshold u increases, ie $\lim_{u \rightarrow \infty} F_u(x) = G(x)$.

The generalised Pareto distribution is a two-parameter distribution with CDF:

$$G(x) = \begin{cases} 1 - \left(1 + \frac{x}{\gamma\beta}\right)^{-\gamma} & \gamma \neq 0 \\ 1 - \exp\left(-\frac{x}{\beta}\right) & \gamma = 0 \end{cases}$$

This distribution has two parameters:

- a scale parameter $\beta > 0$
- a shape parameter γ .

When $\gamma = 0$, this distribution reduces to the exponential distribution.

When $\gamma = 0$, the CDF is:

$$G(x) = 1 - \exp\left(-\frac{x}{\beta}\right)$$

which is the CDF of the $\text{Exp}\left(\frac{1}{\beta}\right)$ distribution.

When $\gamma > 0$, the CDF is:

$$G(x) = 1 - \left(1 + \frac{x}{\gamma\beta}\right)^{-\gamma} = 1 - \left(\frac{\gamma\beta + x}{\gamma\beta}\right)^{-\gamma} = 1 - \left(\frac{\gamma\beta}{\gamma\beta + x}\right)^{\gamma}$$

which is the CDF of the *Pareto* $(\gamma, \gamma\beta)$ distribution.



Question

Derive the PDF for the generalised Pareto distribution.

Solution

In the case when $\gamma \neq 0$:

$$G(x) = 1 - \left(1 + \frac{x}{\gamma\beta}\right)^{-\gamma}$$

Let $v = 1 + \frac{x}{\gamma\beta}$ so that $G(x) = 1 - v^{-\gamma}$. Then:

$$\frac{dv}{dx} = \frac{1}{\gamma\beta} \quad \text{and} \quad \frac{dG(x)}{dv} = \gamma v^{-\gamma-1} = \gamma v^{-(\gamma+1)} = \gamma \left(1 + \frac{x}{\gamma\beta}\right)^{-(\gamma+1)}$$

So the PDF is:

$$\begin{aligned} g(x) &= \frac{dG(x)}{dx} \\ &= \frac{dG(x)}{dv} \times \frac{dv}{dx} \\ &= \gamma \left(1 + \frac{x}{\gamma\beta}\right)^{-\gamma-1} \times \frac{1}{\gamma\beta} \\ &= \frac{1}{\beta} \left(1 + \frac{x}{\gamma\beta}\right)^{-(\gamma+1)} \end{aligned}$$

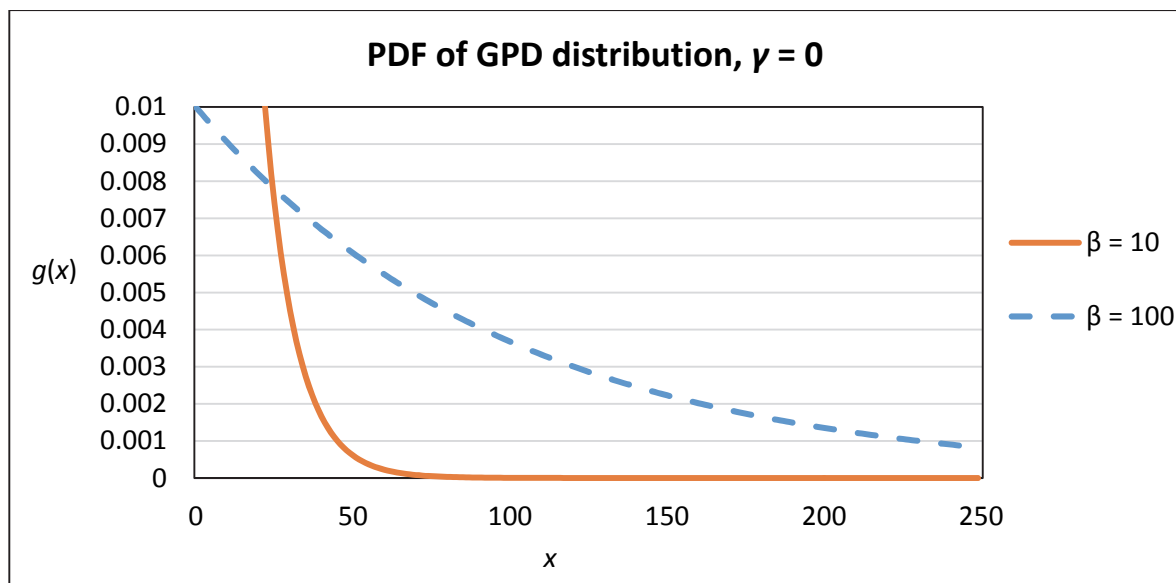
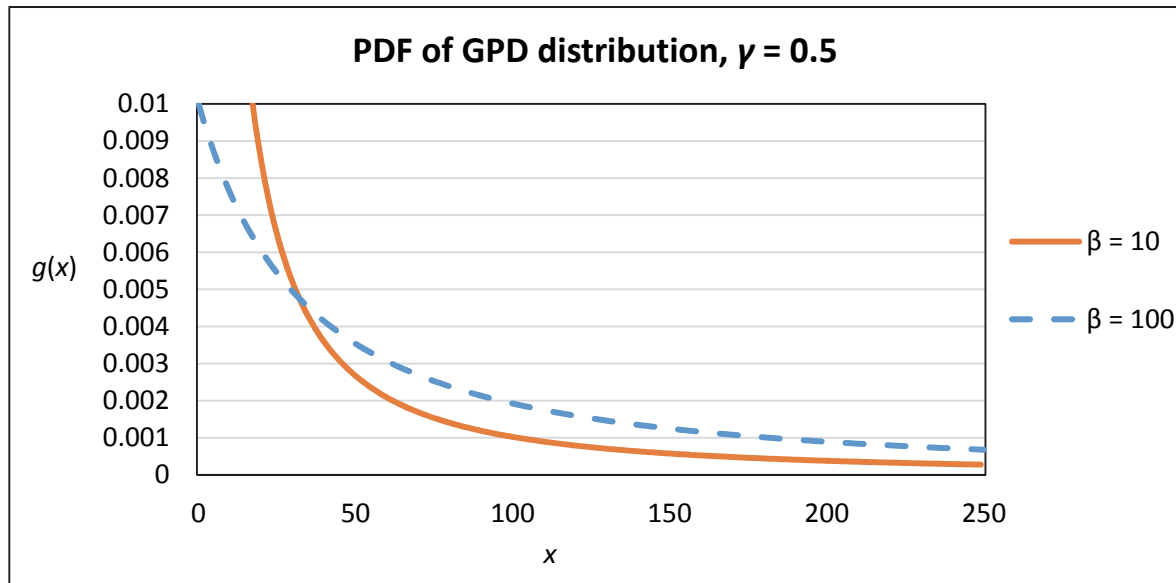
In the case when $\gamma = 0$:

$$G(x) = 1 - \exp\left(-\frac{x}{\beta}\right)$$

Differentiating with respect to x , we see that the PDF is:

$$g(x) = \frac{1}{\beta} \exp\left(-\frac{x}{\beta}\right)$$

The two graphs below illustrate the PDF of the generalised Pareto distribution, using different values for the scale parameter, β , and the shape parameter, γ .



4 Measures of tail weight

There are a number of measures we can use to quantify the *tail weight* of a particular distribution, ie how likely very large values are to occur.

Depending on the context, an exponential, normal or lognormal distribution may be considered to be a suitable baseline to use for comparison.

Tail weight is a measure of how quickly the (upper) tail of a PDF tends to 0. If the PDF of random variable, X_A , tends to 0 as $x \rightarrow \infty$ more slowly than the PDF of random variable, X_B , then X_A is said to have a *heavier* tail than X_B . We will consider four ways of measuring tail weight:

1. the existence of moments
2. limiting density ratios
3. the hazard rate
4. the mean residual life.

4.1 Existence of moments

Recall that the k th moment of a continuous positive-valued distribution with density function $f(x)$ is:

$$\int_0^{\infty} x^k f(x) dx$$

The quantity that is being determined here is the k th non-central moment, $E(X^k)$.

In order for the k th moment to exist, then the integral expression above must converge (ie take a finite value).

For example, for the gamma distribution with density function:

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$$

the k th moment exists for all values of k , indicating that it has a relatively light tail.



Question

Derive a formula for $E(X^k)$ for $k=1,2,3,\dots$ when $X \sim \text{Gamma}(\alpha, \lambda)$.

Solution

If $X \sim \text{Gamma}(\alpha, \lambda)$, then:

$$E(X^k) = \int_0^{\infty} x^k \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx = \int_0^{\infty} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{k+\alpha-1} e^{-\lambda x} dx$$

The integrand can be transformed into the PDF of a $\text{Gamma}(k + \alpha, \lambda)$ distribution by adjusting terms inside and outside of the integral as follows:

$$E(X^k) = \frac{\Gamma(k + \alpha)}{\Gamma(\alpha)\lambda^k} \int_0^{\infty} \frac{\lambda^{k+\alpha}}{\Gamma(k + \alpha)} x^{k+\alpha-1} e^{-\lambda x} dx$$

Since the integral of a PDF over all values of x is equal to 1, the expression reduces to:

$$E(X^k) = \frac{\Gamma(k + \alpha)}{\Gamma(\alpha)\lambda^k} \text{ for } k = 1, 2, 3, \dots$$

(This expression is given on page 12 of the *Tables*.)

The moments can also be obtained from the moment generating function by differentiation.

However, for some distributions, the value of the k th moment does not exist beyond a certain value of k (ie its value becomes infinite).

For the Pareto distribution with density function:

$$f(x) = \frac{\alpha\lambda^\alpha}{(\lambda + x)^{\alpha+1}}$$

the k th moment only exists when $k < \alpha$.

So a Pareto distribution (with a low value of the parameter α) will have a much thicker tail.

For $X \sim \text{Pa}(\alpha, \lambda)$, the mean and variance are given in the *Tables* as follows:

$$E(X) = \frac{\lambda}{\alpha - 1} \quad \text{and} \quad \text{var}(X) = \frac{\alpha\lambda^2}{(\alpha - 1)^2(\alpha - 2)}$$

From the denominators of these two expressions, we see that the mean is defined only for $\alpha > 1$ and the variance is defined only for $\alpha > 2$.

4.2 Limiting density ratios

We can compare the thickness of the tail of two distributions by calculating the relative values of their density functions at the far end of the upper tail. For example, if we compare the Pareto distributions with parameters $\alpha = 2$ and $\alpha = 3$ (both with the same value of λ), we find that:

$$\lim_{x \rightarrow \infty} \frac{f_{\alpha=2}(x)}{f_{\alpha=3}(x)} = \lim_{x \rightarrow \infty} \left\{ \frac{2\lambda^2}{(\lambda + x)^3} \bigg/ \frac{3\lambda^3}{(\lambda + x)^4} \right\} = \frac{2}{3\lambda} \lim_{x \rightarrow \infty} (\lambda + x) = \infty$$

This confirms that the distribution with $\alpha = 2$ has a much thicker tail.

If we compare the gamma distribution with the Pareto distribution, we find that the presence of the exponential factor in the gamma density results in a limiting density ratio of zero, which confirms that the gamma distribution has a lighter tail.



Question

Consider the $\text{Gamma}(0.5, 0.005)$ and $\text{Pa}(4, 300)$ distributions, both of which have a mean of 100 and a variance of 20,000.

Use the 'limiting density ratio' method to compare these two distributions.

Solution

Comparing the two density functions and taking the limit as $x \rightarrow \infty$, we have:

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{f_{\text{gamma}}(x)}{f_{\text{pareto}}(x)} &= \lim_{x \rightarrow \infty} \left\{ \left(\frac{0.005^{0.5}}{\Gamma(0.5)} x^{-0.5} e^{-0.005x} \right) \middle/ \left(\frac{4 \times 300^4}{(300+x)^5} \right) \right\} \\ &= \lim_{x \rightarrow \infty} \left\{ C x^{-0.5} (300+x)^5 e^{-0.005x} \right\} \end{aligned}$$

In the above expression, C is a (very small) constant. (We know that $\Gamma(0.5) = \sqrt{\pi}$ from the properties of the gamma function given on page 5 of the *Tables*.)

As $x \rightarrow \infty$, the $e^{-0.005x}$ factor tends to 0, the $x^{-0.5}$ factor tends to 0 and the $(300+x)^5$ factor tends to infinity, and so the overall expression tends to 0. (To see, this, try some large values of x .) Therefore:

$$\lim_{x \rightarrow \infty} \frac{f_{\text{gamma}}(x)}{f_{\text{pareto}}(x)} = 0$$

This suggests that the gamma distribution has a lighter tail than the Pareto distribution.



We can obtain the values of the PDF of two distributions x_1 and x_2 for, say, x values 1 to 1,000 and then calculate the ratio, R , using:

$$R = x_1/x_2$$

We can then plot the graph of R against x to determine which of x_1 and x_2 has the thicker tail.

4.3 Hazard rate

The **hazard rate** of a distribution with density function $f(x)$ and distribution function $F(x)$ is defined as:

$$h(x) = \frac{f(x)}{1 - F(x)}$$

We have already seen this formula in [Chapter 8](#). Recall that the hazard rate is the rate of failure given survival up until that point.



Question

- (i) Determine the hazard rate for:
- (a) the $\text{Exp}(\lambda)$ distribution
 - (b) the $\text{Pa}(\alpha, \lambda)$ distribution.
- (ii) Comment on the differences between these hazard rates.

Solution

- (i)(a) The hazard rate for the exponential distribution is:

$$h(x) = \frac{f(x)}{1 - F(x)} = \frac{\lambda e^{-\lambda x}}{e^{-\lambda x}} = \lambda$$

- (i)(b) The hazard rate for the Pareto distribution is:

$$h(x) = \frac{f(x)}{1 - F(x)} = \frac{\alpha \lambda^\alpha}{(\lambda + x)^{\alpha+1}} \bigg/ \left(\frac{\lambda}{\lambda + x} \right)^\alpha = \frac{\alpha}{\lambda + x}$$

- (ii) The exponential hazard rate is constant (*ie* it is independent of x , which demonstrates the memoryless property of the exponential distribution), whereas the Pareto hazard rate is a decreasing function of x .

We can interpret the hazard rate by analogy with μ_x , the force of mortality at age x .

The force of mortality has been discussed in detail in earlier chapters.

(The force of mortality at age x ($0 \leq x < \omega$) is defined as:

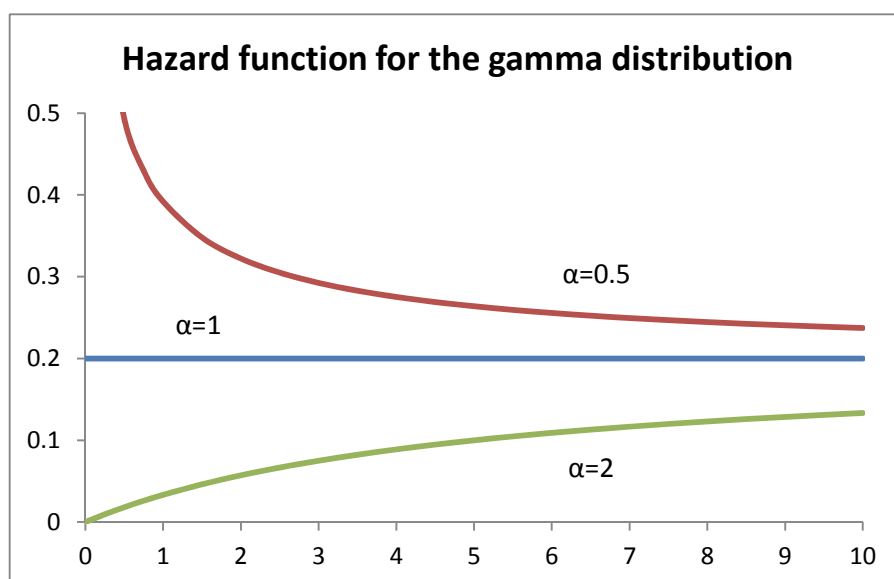
$$\mu_x = \lim_{h \rightarrow 0^+} \frac{1}{h} \times P[T \leq x + h \mid T > x]$$

where T is a random variable measuring a person's length of life. See [Chapter 6, Section 1.3.](#))

If the force of mortality increases as a person's age increases, relatively few people will live to old age (corresponding to a light tail). If, on the other hand, the force of mortality decreases as the person's age increases, there is the potential to live to a very old age (corresponding to a heavier tail).

For the gamma distribution we find that, if $\alpha = 1$ (*ie* it is an exponential distribution), the hazard rate is constant, but if $\alpha < 1$, it is a decreasing function of x (indicating a heavier tail than the exponential distribution) and if $\alpha > 1$, it is an increasing function (indicating a lighter tail than the exponential distribution).

The graph below illustrates the hazard function for the $\text{Gamma}(\alpha, \lambda)$ distribution for different values of α and $\lambda = 0.2$. The axes of the graph are the hazard function $h(x) = \frac{f(x)}{1-F(x)}$ against x .



The formula for hazard function in each of the three cases is:

- $$h(x) = \frac{f(x)}{1-F(x)} = \frac{\frac{\lambda^2}{\Gamma(2)} x^{2-1} e^{-\lambda x}}{1 - \int_0^x \frac{\lambda^2}{\Gamma(2)} t^{2-1} e^{-\lambda t} dt} = \frac{\lambda^2 x e^{-\lambda x}}{1 - \int_0^x \lambda^2 t e^{-\lambda t} dt} \text{ when } \alpha = 2$$
- $$h(x) = \frac{f(x)}{1-F(x)} = \frac{\frac{\lambda}{\Gamma(1)} x^{1-1} e^{-\lambda x}}{1 - \int_0^x \frac{\lambda}{\Gamma(1)} t^{1-1} e^{-\lambda t} dt} = \frac{\lambda e^{-\lambda x}}{1 - \int_0^x \lambda e^{-\lambda t} dt} \text{ when } \alpha = 1$$
- $$h(x) = \frac{f(x)}{1-F(x)} = \frac{\frac{\lambda^{0.5}}{\Gamma(0.5)} x^{0.5-1} e^{-\lambda x}}{1 - \int_0^x \frac{\lambda^{0.5}}{\Gamma(0.5)} t^{0.5-1} e^{-\lambda t} dt} = \frac{\frac{\lambda^{0.5}}{\sqrt{\pi}} x^{-0.5} e^{-\lambda x}}{1 - \int_0^x \frac{\lambda^{0.5}}{\sqrt{\pi}} t^{-0.5} e^{-\lambda t} dt} \text{ when } \alpha = 0.5.$$

For the Pareto distribution, we find that the hazard rate is always a decreasing function of x , confirming that it has a heavy tail.

We derived these results for the exponential and Pareto distributions in the previous question.



The R code to calculate the hazard rate, H , for a Weibull distribution with parameters g and $b = c^{-1/g}$ is given by:

```
H<-dweibull(x,g,b)/(1-pweibull(x,g,b))
```

We can then plot the graph of H against x to determine the thickness of its tail.

4.4 Mean residual life

The *mean residual life* of a distribution with density function $f(x)$ and distribution function $F(x)$ is defined as:

$$e(x) = \frac{\int_x^\infty (y-x)f(y)dy}{\int_x^\infty f(y)dy} = \frac{\int_x^\infty \{1-F(y)\}dy}{1-F(x)}$$

This function gives the expected remaining survival time given survival up until this point.



Question

- (i) Determine the mean residual life for:
- the $\text{Exp}(\lambda)$ distribution
 - the $\text{Pa}(\alpha, \lambda)$ distribution where $\alpha > 1$.
- (ii) Comment on the behaviour of these functions.

Solution

- (i)(a) The mean residual life for the exponential distribution is:

$$\begin{aligned} e(x) &= \frac{\int_x^\infty \{1-F(y)\}dy}{1-F(x)} = \frac{\int_x^\infty e^{-\lambda y} dy}{e^{-\lambda x}} \\ &= \frac{\left[-\frac{1}{\lambda} e^{-\lambda y} \right]_x^\infty}{e^{-\lambda x}} = \frac{1}{\lambda} \end{aligned}$$

- (i)(a) The mean residual life for the Pareto distribution with $\alpha > 1$ is:

$$\begin{aligned} e(x) &= \frac{\int_x^\infty \{1-F(y)\}dy}{1-F(x)} = \frac{\int_x^\infty \left(\frac{\lambda}{\lambda+y} \right)^\alpha dy}{\left(\frac{\lambda}{\lambda+x} \right)^\alpha} = (\lambda+x)^\alpha \int_x^\infty (\lambda+y)^{-\alpha} dy \\ &= (\lambda+x)^\alpha \left[\frac{1}{-\alpha+1} (\lambda+y)^{-\alpha+1} \right]_x^\infty = (\lambda+x)^\alpha \left[0 - \left(\frac{1}{-\alpha+1} \right) (\lambda+x)^{-\alpha+1} \right] \\ &= \frac{\lambda+x}{\alpha-1} \end{aligned}$$

- (ii) The exponential mean residual life is constant whereas the Pareto mean residual life is an increasing function of x .

When $x = 0$ the mean residual life is equal to the mean of the underlying distribution, as we would expect.

Again we can interpret this in terms of mortality as the expected future lifetime, e_x .

We discussed expected future lifetimes in [Chapter 6](#). There we derived the formula:

$$e_x = \int_0^{\infty} {}_t p_x dt$$

However since:

$${}_t p_x = \frac{{}_{t+x} p_0}{{}_x p_0} = \frac{S(t+x)}{S(x)} = \frac{1-F(t+x)}{1-F(x)}$$

we have:

$$e_x = \frac{\int_0^{\infty} \{1-F(t+x)\} dt}{1-F(x)}$$

Then making the substitution $y = t + x$, we see that:

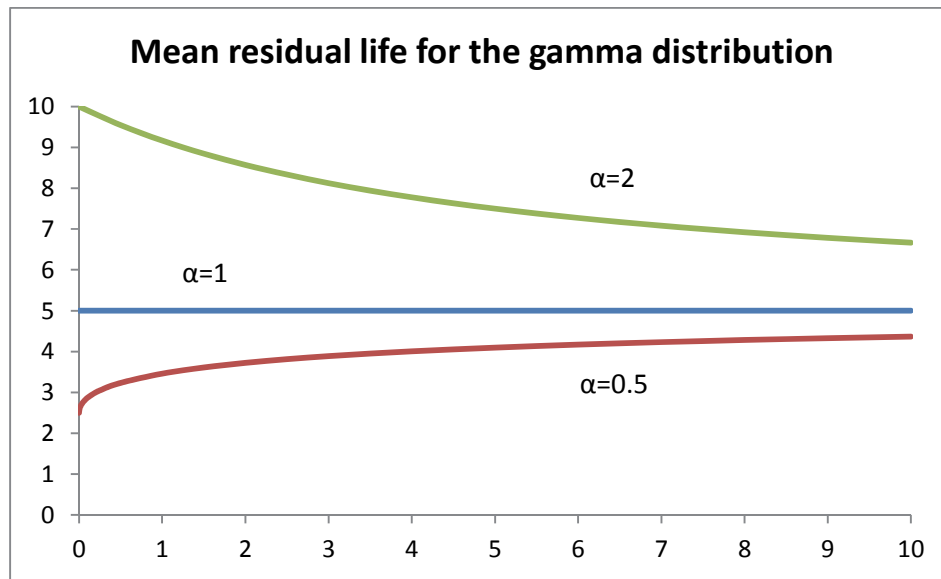
$$e_x = \frac{\int_x^{\infty} \{1-F(y)\} dy}{1-F(x)} = e(x)$$

If the expected future lifetime decreases with age, relatively few people will live to old age (corresponding to a light tail), but if it increases, there is the potential to live to a very old age (corresponding to a heavier tail).

For the gamma distribution we find that, if $\alpha = 1$ (ie it is an exponential distribution), the mean residual life is constant, but if $\alpha < 1$, it is an increasing function of x (indicating a heavier tail than the exponential distribution) and if $\alpha > 1$, it is a decreasing function (indicating a lighter tail than the exponential distribution).

The graph below illustrates the mean residual life for the $\text{Gamma}(\alpha, \lambda)$ distribution for different

values of α and $\lambda = 0.2$. The axes of the graph are the mean residual life $e(x) = \frac{\int_x^{\infty} \{1-F(y)\} dy}{1-F(x)}$ against x .



For the Pareto distribution, we find that the mean residual life is always an increasing function of x , confirming that it has a heavy tail.

We derived these results for the exponential and Pareto distributions in the previous question.



The R code for the survival function of a Weibull distribution with parameters g and $b = c^{-1/g}$ is given by:

```
Sy<-function(y) {(1-pweibull(y,g,b))}
```

Hence, the mean residual life for x is given by e_x as follows:

```
int<-integrate(Sy,x,Inf)
ex<-int$value/(1-pweibull(x,g,b))
```

We can then plot the graph of e_x against x .

Chapter 16 Summary

Extreme events

An extreme event is one with a very low frequency and very high severity.

Modelling extreme financial events is difficult due to:

- lack of historic data in the tails of the distribution
- the 'true' distribution of many types of financial data being more leptokurtic (more peaked with fatter tails) than the normal distribution
- the volatility of financial variables being heteroscedastic (varying over time)
- parameter estimates being inappropriately influenced by the main bulk of the data in the middle of the distribution.

Better modelling of the tails of the data can be done using extreme value theory.

Extreme value theory

Extreme value theory attempts to model the asymptotic behaviour of the tails of distributions. There are two main approaches:

1. modelling the maximum values of a distribution – using the generalised extreme value family of distributions
2. modelling the values exceeding a threshold – using the generalised Pareto family of distributions.

Generalised extreme value distributions

Let:

- losses X_i be IID with cumulative distribution $F(x_i)$
- $X_M = \max\{X_1, X_2, \dots, X_n\}$ be the block maxima
- $\alpha_1, \dots, \alpha_n$ and $\beta_1, \dots, \beta_n > 0$ be suitable sequences of real constants.

Then, if n is sufficiently large, the distribution of the standardised block maxima, $\frac{X_M - \alpha_n}{\beta_n}$,

is asymptotically described by the generalised extreme value (GEV) family of distributions with CDF:

$$H(x) = \lim_{n \rightarrow \infty} P\left(\frac{X_M - \alpha_n}{\beta_n} \leq x\right) = \lim_{n \rightarrow \infty} [F(\beta_n x + \alpha_n)]^n$$

The cumulative distribution function of the GEV distribution is:

$$H(x) = \begin{cases} \exp\left(-\left(1 + \frac{\gamma(x-\alpha)}{\beta}\right)^{-1/\gamma}\right) & \gamma \neq 0 \\ \exp\left(-\exp\left(-\frac{(x-\alpha)}{\beta}\right)\right) & \gamma = 0 \end{cases}$$

The three parameters of the GEV family are:

- a location parameter, α
- a scale parameter, $\beta > 0$
- a shape parameter, γ .

There are three types of GEV distributions, which are distinguished by the sign of the shape parameter γ . Each type corresponds to different underlying loss distributions:

GEV distributions (for the maximum value) corresponding to common loss distributions			
Type	WEIBULL	GUMBEL	FRÉCHET
Shape parameter	$\gamma < 0$	$\gamma = 0$	$\gamma > 0$
Underlying distribution	Beta Uniform Triangular	Chi-square Exponential Gamma Lognormal Normal Weibull	Burr F Log-gamma Pareto t
Range of values permitted	$x < \alpha - \frac{\beta}{\gamma}$	$-\infty < x < \infty$	$x > \alpha - \frac{\beta}{\gamma}$

Generalised Pareto distributions

Let losses X_i be IID with cumulative distribution $F(x_i)$. Then the distribution of the conditional losses above a threshold, u , will converge (whatever the underlying distribution of the data) to a generalised Pareto distribution (GPD) with CDF:

$$G(x) = \lim_{u \rightarrow \infty} F_u(x) = \lim_{u \rightarrow \infty} P(X - u \leq x | X > u) = \lim_{u \rightarrow \infty} \frac{F(x + u) - F(u)}{1 - F(u)}$$

The CDF of the GPD is of the form:

$$G(x) = \begin{cases} 1 - \left(1 + \frac{x}{\gamma\beta}\right)^{-\gamma} & \gamma \neq 0 \\ 1 - \exp\left(-\frac{x}{\beta}\right) & \gamma = 0 \end{cases}$$

The two parameters of the GPD family are:

- a scale parameter, $\beta > 0$
- a shape parameter, γ .

Measures of tail weight

There are a number of measures we can use to quantify the tail weight of a particular distribution, ie how likely very large values are to occur:

- the existence of moments
 - The existence of all moments, $E(X^k)$, for all positive integers, k , indicates a light tail. If moments exist only up to a positive integer, k , this is an indication that the distribution has a heavy tail.
- limiting density ratios
 - The limiting value, as $x \rightarrow \infty$, of the ratio of two PDFs can be used to determine which distribution has the lighter or heavier tail.
- the hazard rate
 - An increasing (decreasing) hazard rate, $h(x) = \frac{f(x)}{1 - F(x)}$, corresponds to a lighter (heavier) tail.
- the mean residual life
 - An increasing (decreasing) mean residual life, $e(x) = \frac{\int_x^\infty \{1 - F(y)\} dy}{1 - F(x)}$, corresponds to a heavier (lighter) tail.

The practice questions start on the next page so that you can keep the chapter summaries together for revision purposes.



Chapter 16 Practice Questions

16.1 (i) Explain what is meant by an extreme event and give two examples in an insurance context.

(ii) Explain why it is important to model extreme events separately from other events.

16.2 (i) Describe the generalised extreme value (GEV) distribution. [7]

Exam style

(ii) Outline an alternative approach that can be used in place of the GEV distribution to model extreme events. [3]

(iii) State the key advantage of the method outlined in (ii) over that described in (i). [1]
[Total 11]

16.3 The claim amounts in a general insurance portfolio are independent and follow an exponential distribution with mean £2,500.

Exam style

(i) Calculate the probability that an individual claim will exceed £10,000. [1]

(ii) Calculate the probability that, in a sample of 100 claims, the largest claim will exceed £10,000 using:

(a) an exact method

(b) an approximation based on a Gumbel-type GEV distribution. [5]

You are given that, for an exponential distribution with parameter λ , the approximate distribution of $\max\{X_1, \dots, X_n\}$ for large n is a Gumbel-type GEV distribution with CDF:

$$H(x) = \exp\left(-\exp\left(-\left(\frac{x - \alpha_n}{\beta_n}\right)\right)\right) \text{ where } \alpha_n = \frac{1}{\lambda} \ln n \text{ and } \beta_n = \frac{1}{\lambda}$$

(iii) State the two key assumptions made in (ii)(a). [1]

[Total 7]

16.4 If individual losses, X , follow a $Pa(\alpha, \lambda)$ distribution, determine the distribution of the threshold exceedances, $W = X - u \mid X \geq u$.

16.5 Compare the limiting value of the density functions for a $Gamma(\alpha, \lambda)$ and an $Exp(\lambda)$ distribution when $\alpha > 1$ and hence determine which has the heavier tail.

16.6 (i) Determine the hazard rate for the Weibull distribution with parameters $c > 0$ and $\gamma > 0$.

(ii) Comment on the behaviour of the hazard rate.

16.7 (i) Show that:

Exam style

$$\int_x^\infty e^{-3y^2} dy = \frac{2}{9} P\left(\chi_4^2 > 6x^{\frac{1}{2}}\right) \quad [5]$$

Hint: use the substitution $u = 3y^2$ and transform the integrand into the PDF of the $\text{Gamma}(2,1)$ distribution.

(ii) Hence deduce an expression involving a chi-squared probability for the mean residual life for the $W\left(3, \frac{1}{2}\right)$ distribution. [2]

(iii) By calculating the values of mean residual life function when $x = 1$, $x = 4$ and $x = 9$, determine whether the mean residual life of the $W\left(3, \frac{1}{2}\right)$ distribution is an increasing or decreasing function of x . [2]

[Total 9]



Chapter 16 Solutions

16.1 (i) *Extreme events*

Extreme events are outcomes that have a very low probability of occurrence but involve very large sums of money.

In an insurance context, they may arise as a result of a single cause that has a high financial cost (*eg* a bodily injury claim or complete destruction of a building) ...

... or as an accumulation of events with a related cause (*eg* flood damage to a large number of houses in one town).

(ii) *Why extreme events are modelled separately*

The majority of risk events fall within the main body of the fitted distribution and can usually be modelled reasonably accurately by one of the standard statistical distributions.

However, there is usually a lack of past data on extreme events.

If a distribution is fitted to the whole dataset, the parameter estimates will reflect where the bulk of the data values lie rather than the extreme events. This might mean the fitted distribution understates the probability of future extreme events.

Therefore, a different approach to modelling extreme events is taken, *eg* by considering the distribution of block maxima or the distribution of threshold exceedances.

16.2 (i) *Describe the GEV distribution*

The maximum value, X_M , in a sample of n IID random variables $X_1, X_2, \dots, X_n$ tends to a particular distribution as the sample size increases. This is called the generalised extreme value (GEV) distribution. [1]

The GEV distribution has CDF:

$$H(x) = \begin{cases} \exp\left(-\left(1 + \frac{\gamma(x-\alpha)}{\beta}\right)^{-1/\gamma}\right) & \gamma \neq 0 \\ \exp\left(-\exp\left(-\frac{(x-\alpha)}{\beta}\right)\right) & \gamma = 0 \end{cases} \quad [1]$$

The key parameter is the shape parameter, γ . [½]

When $\gamma > 0$, we have the Fréchet-type GEV distribution. [½]

This is the limiting form for 'heavy tail' underlying distributions with a finite lower bound, such as the Pareto distribution. [1]

When $\gamma < 0$, we have the Weibull-type GEV distribution. [½]

This is the limiting form for underlying distributions with a finite upper bound, such as the uniform distribution. [1]

When $\gamma = 0$, we have the Gumbel-type GEV distribution. [½]

This is the limiting form for most other underlying distributions that have finite moments, such as the normal and lognormal distributions. [½]

The parameters α and β are the location and scale parameters, respectively. These will differ depending on the underlying distribution. [½]

(ii) **Alternative approach**

As an alternative to focusing upon a single maximum value, we can consider the distribution of *all* the claim values that exceed some threshold, u . The distribution of $X - u \mid X > u$ is called the threshold exceedances distribution. [1]

A similar theory to GEV predicts that the limiting distribution, as $u \rightarrow \infty$, is a generalised Pareto distribution (GPD). [½]

The GPD has CDF:

$$G(x) = \begin{cases} 1 - \left(1 + \frac{x}{\gamma\beta}\right)^{-\gamma} & \gamma \neq 0 \\ 1 - \exp\left(-\frac{x}{\beta}\right) & \gamma = 0 \end{cases} \quad [1]$$

The parameters β and γ are the scale and shape parameters, respectively.

In order to fit the tail of a distribution we need to select a suitably high threshold and then fit the GPD to the values in excess of that threshold. [½]

(iii) **Key advantage of the GPD method**

The GPD method has the advantage that it uses a larger part of the data and models *all* the large claims above the threshold, not just the single highest value. [1]

16.3 (i) **Probability of a claim greater than £10,000 using an exponential distribution**

The claims distribution is $\text{Exp}\left(\frac{1}{2,500}\right)$.

Using the formula for CDF of an exponential random variable given on page 11 of the *Tables*, we have:

$$P(X > 10,000) = 1 - P(X \leq 10,000) = 1 - F_X(10,000) = e^{-10,000\lambda} = e^{-4} = 0.0183 \quad [1]$$

(ii)(a) **Probability that at least one claim will exceed £10,000 using an exact method**

The required probability is:

$$\begin{aligned}
 P(\max\{X_1, \dots, X_{100}\} > 10,000) &= 1 - P(\max\{X_1, \dots, X_{100}\} \leq 10,000) \\
 &= 1 - P(X_1 \leq 10,000, \dots, X_{100} \leq 10,000) \\
 &= 1 - P(X_1 \leq 10,000) \times \dots \times P(X_{100} \leq 10,000) \\
 &= 1 - (1 - e^{-4})^{100} = 0.8425
 \end{aligned}
 \tag{2}$$

(ii)(b) **Probability that at least one claim will exceed £10,000 using an approximate method**

The approximate distribution of $\max\{X_1, \dots, X_{100}\}$ is a Gumbel-type GEV distribution with CDF:

$$H(x) = \exp\left(-\exp\left(-\left(\frac{x - \alpha_{100}}{\beta_{100}}\right)\right)\right)$$

where:

$$\alpha_{100} = 2,500 \ln 100 = 11,512.93 \quad \text{and} \quad \beta_{100} = 2,500 \tag{1}$$

So:

$$\begin{aligned}
 P(\max\{X_1, \dots, X_{100}\} > 10,000) &= 1 - P(\max\{X_1, \dots, X_{100}\} \leq 10,000) \\
 &\approx 1 - \exp\left(-\exp\left(-\left(\frac{10,000 - \alpha_{100}}{\beta_{100}}\right)\right)\right) \\
 &= 1 - \exp\left(-\exp\left(-\left(\frac{10,000 - 11,512.93}{2,500}\right)\right)\right) \\
 &= 1 - 0.1602 = 0.8398
 \end{aligned}
 \tag{2}$$

(iii) **Assumptions**

The two key assumptions are that all claims come from an exponential distribution with mean £2,500 and that they are statistically independent. [1]

16.4 The CDF of the threshold exceedances is given by:

$$F_u(x) = \frac{F(x+u) - F(u)}{1 - F(u)}$$

If the individual losses follow a $Pa(\alpha, \lambda)$ distribution, then:

$$\begin{aligned} F_u(x) &= \frac{\left\{1 - \left(\frac{\lambda}{\lambda + x + u}\right)^\alpha\right\} - \left\{1 - \left(\frac{\lambda}{\lambda + u}\right)^\alpha\right\}}{1 - \left\{1 - \left(\frac{\lambda}{\lambda + u}\right)^\alpha\right\}} \\ &= \frac{\left(\frac{\lambda}{\lambda + u}\right)^\alpha - \left(\frac{\lambda}{\lambda + x + u}\right)^\alpha}{\left(\frac{\lambda}{\lambda + u}\right)^\alpha} \\ &= 1 - \left(\frac{\lambda + u}{\lambda + u + x}\right)^\alpha \end{aligned}$$

This is the CDF of the $Pa(\alpha, \lambda + u)$ distribution. So the distribution of the threshold exceedances is $Pa(\alpha, \lambda + u)$.

16.5 Comparing the two density functions and taking the limit as $x \rightarrow \infty$, we have:

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{f_{gamma}(x)}{f_{exp}(x)} &= \lim_{x \rightarrow \infty} \left\{ \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} / \lambda e^{-\lambda x} \right\} \\ &= \lim_{x \rightarrow \infty} \left\{ C x^{\alpha-1} \right\} \end{aligned}$$

where C is a constant.

When $\alpha > 1$:

$$\frac{f_{gamma}(x)}{f_{exp}(x)} \rightarrow \infty \text{ as } x \rightarrow \infty$$

Hence the $Gamma(\alpha, \lambda)$ distribution has the heavier tail.

16.6 (i) Hazard rate for the Weibull distribution

The hazard rate for the $W(c, \gamma)$ distribution is:

$$\begin{aligned} h(x) &= \frac{f(x)}{1-F(x)} \\ &= \frac{c\gamma x^{\gamma-1} e^{-cx^\gamma}}{e^{-cx^\gamma}} \\ &= c\gamma x^{\gamma-1} \end{aligned}$$

(ii) Comment on behaviour

If $\gamma > 1$, then this hazard rate is an increasing function of x , which corresponds to a light tail.

If $0 < \gamma < 1$, then the hazard rate is a decreasing function of x , which corresponds to a heavy tail.

16.7 (i) Show the integral expression

Using the substitution, $u = 3y^{\frac{1}{2}}$, we have:

$$y = \left(\frac{u}{3}\right)^2$$

and $\frac{dy}{du} = \frac{2}{9}u$ [1]

Hence the integral becomes:

$$\int_x^\infty e^{-3y^{\frac{1}{2}}} dy = \int_{3x^{\frac{1}{2}}}^\infty \frac{2}{9} u e^{-u} du \quad [1]$$

The integrand now resembles, but is not quite, that of the $Gamma(2,1)$ distribution.

We can transform the integrand into that of the $Gamma(2,1)$ distribution by adjusting the constants inside and outside the integral:

$$\int_x^\infty e^{-3y^{\frac{1}{2}}} dy = \frac{2}{9} \left(\frac{\Gamma(2)}{1^2} \right) \int_{3x^{\frac{1}{2}}}^\infty \frac{1^2}{\Gamma(2)} u e^{-u} du \quad [1]$$

Now $\Gamma(2) = 1! = 1$ and the integral is the probability that a $Gamma(2,1)$ random variable takes a value greater than $3x^{\frac{1}{2}}$. So:

$$\int_x^\infty e^{-3y^{\frac{1}{2}}} dy = \frac{2}{9} P\left(U > 3x^{\frac{1}{2}}\right) \quad \text{where } U \sim Gamma(2,1) \quad [1]$$

Using the relationship between the gamma and chi-squared distributions (given on page 12 of the *Tables*), we see that that $2U \sim \chi_4^2$. Hence:

$$\int_x^\infty e^{-3y^2} dy = \frac{2}{9} P\left(2U > 6x^{\frac{1}{2}}\right) = \frac{2}{9} P\left(\chi_4^2 > 6x^{\frac{1}{2}}\right) \quad [1]$$

(ii) **Mean residual life**

The mean residual life for the *Weibull* $\left(3, \frac{1}{2}\right)$ distribution is:

$$\begin{aligned} e(x) &= \frac{\int_x^\infty \{1 - F(y)\} dy}{1 - F(x)} \\ &= \frac{\int_x^\infty e^{-3y^2} dy}{e^{-3x^2}} \\ &= \frac{\frac{2}{9} P\left(\chi_4^2 > 6x^{\frac{1}{2}}\right)}{e^{-3x^2}} \end{aligned} \quad [2]$$

(iii) **Nature of the mean residual life function**

$$\text{When } x = 1, \text{ we have } e(x) = \frac{\frac{2}{9} P\left(\chi_4^2 > 6\right)}{e^{-3}} = \frac{\frac{2}{9} \times (1 - 0.8009)}{e^{-3}} = 0.8887. \quad [\frac{1}{2}]$$

$$\text{When } x = 4, \text{ we have } e(x) = \frac{\frac{2}{9} P\left(\chi_4^2 > 12\right)}{e^{-6}} = \frac{\frac{2}{9} \times (1 - 0.9826)}{e^{-6}} = 1.5599. \quad [\frac{1}{2}]$$

$$\text{When } x = 9, \text{ we have } e(x) = \frac{\frac{2}{9} P\left(\chi_4^2 > 18\right)}{e^{-9}} = \frac{\frac{2}{9} \times (1 - 0.9988)}{e^{-9}} = 2.1608. \quad [\frac{1}{2}]$$

The mean residual life is an increasing function of x , suggesting that this distribution has a heavy tail. [½]

End of Part 4

What next?

1. Briefly **review** the key areas of Part 4 and/or re-read the **summaries** at the end of Chapters **13** to **16**.
2. Ensure you have attempted some of the **Practice Questions** at the end of each chapter in Part 4. If you don't have time to do them all, you could save the remainder for use as part of your revision.
3. Attempt **Assignment X4**.

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